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# =====
# PW-GWAS-informed colocalisation analysis
#
# Example: Frailty Index (FI) vs Rheumatoid Arthritis (RA)
#
# PW-GWAS interpretation:
#   Model 3 = evidence for a shared causal variant
#   Model 4 = evidence for distinct causal variants
#
# This script:
#   1. Reads PW-GWAS block-level results
#   2. Selects high-confidence Model 3 and Model 4 regions
#   3. Excludes the MHC and APOE regions
#   4. Harmonises GWAS summary statistics
#   5. Performs regional colocalisation using coloc
#
# Outputs:
#   FI_RA_coloc_shared_QC.tsv
#   FI_RA_coloc_distinct_QC.tsv
# =====

suppressPackageStartupMessages({
  library(data.table)
  library(coloc)
})

stopifnot(as.character(packageVersion("coloc")) >= "5.2.3")

# -----
# 1. Sample information
# -----

# Example case-control GWAS sample sizes.
# Replace these values with those corresponding to the outcome GWAS used.
N_cases    <- 14361
N_controls  <- 43923
N_total    <- N_cases + N_controls

# Proportion of cases required by coloc for case-control traits
case_prop <- N_cases / N_total

# -----
# 2. Analysis settings

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TRAIT_1 <- "FI"
TRAIT_2 <- "RA"

# GWAS summary-statistic files
TRAIT1_FILE <- "sumstats_FI_hg19.tsv.gz"
TRAIT2_FILE <- "sumstats_RA_hg19.tsv.gz"

# PW-GWAS block summary
BLOCK_FILE <- if (file.exists("FI_RA_block_summary.tsv")) {
  "FI_RA_block_summary.tsv"
} else if (file.exists("RA_FI_block_summary.tsv")) {
  "RA_FI_block_summary.tsv"
} else {
  stop("No FI-RA PW-GWAS block summary file found.")
}

# Output files
SHARED_OUT <- sprintf(
  "%s_%s_coloc_shared_QC.tsv",
  TRAIT_1,
  TRAIT_2
)

DISTINCT_OUT <- sprintf(
  "%s_%s_coloc_distinct_QC.tsv",
  TRAIT_1,
  TRAIT_2
)

# Padding around each PW-GWAS region
PAD <- 250000L

cat("\n=====\n")
cat(
  "Running PW-GWAS-informed coloc for",
  TRAIT_1,
  "vs",
  TRAIT_2,
  "\n"
)
cat("=====\n")

# -----
# 3. Load PW-GWAS block results

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blocks <- fread(BLOCK_FILE)

# Exclude complex genomic regions:
#   MHC: chr6:25-35 Mb
#   APOE: chr19:44.4-45.9 Mb

blocks_filtered <- blocks[
  !(chr == 6 & st < 35e6 & sp > 25e6) &
  !(chr == 19 & st < 45.9e6 & sp > 44.4e6)
]

# -----
# 4. Identify PW-GWAS Model 3 and Model 4 columns
# -----

P3_col <- intersect(
  c("P3", "PPA_3", "MAP3", "post3", "prob_M3"),
  names(blocks_filtered)
)[1]

P4_col <- intersect(
  c("P4", "PPA_4", "MAP4", "post4", "prob_M4"),
  names(blocks_filtered)
)[1]

if (is.na(P3_col) || is.na(P4_col)) {
  stop("Could not identify Model 3 / Model 4 posterior-probability columns.")
}

# Select high-confidence regions
shared_blocks <- blocks_filtered[
  get(P3_col) >= 0.90,
  unique(block)
]

distinct_blocks <- blocks_filtered[
  get(P4_col) >= 0.90,
  unique(block)
]

cat(
  "Shared blocks (Model 3 >= 0.90):",
  paste(shared_blocks, collapse = ", "),

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    "\n"
  )

cat(
  "Distinct blocks (Model 4 >= 0.90):",
  paste(distinct_blocks, collapse = ", "),
  "\n"
)

# -----
# 5. Helper functions
# -----

# Merge two GWAS datasets using SNP ID where possible.
# Otherwise merge by chromosome and genomic position.

merge_gwas <- function(A, B) {

  if ("snp" %in% names(A) && "snp" %in% names(B)) {

    merge(
      A,
      B,
      by = "snp",
      suffixes = c(".A", ".B")
    )

  } else {

    M <- merge(
      A,
      B,
      by = c("chr", "pos"),
      suffixes = c(".A", ".B")
    )

    if (!"snp" %in% names(M)) {
      M[, snp := paste0(chr, ":", pos)]
    }

    M
  }
}

# Identify strand-ambiguous SNPs

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is_ambiguous <- function(effect_allele, other_allele) {

  allele_pair <- paste0(
    toupper(effect_allele),
    toupper(other_allele)
  )

  allele_pair %in% c(
    "AT",
    "TA",
    "CG",
    "GC"
  )
}

# Harmonise effect alleles across the two GWAS

harmonise_gwas <- function(M) {

  allele_columns <- c(
    "EA.A",
    "OA.A",
    "EA.B",
    "OA.B"
  )

  if (all(allele_columns %in% names(M))) {

    same <- (
      M$EA.A == M$EA.B &
      M$OA.A == M$OA.B
    )

    flipped <- (
      M$EA.A == M$OA.B &
      M$OA.A == M$EA.B
    )

    ambiguous <- (
      is_ambiguous(M$EA.A, M$OA.A) |
      is_ambiguous(M$EA.B, M$OA.B)
    )

    # Reverse the Trait 2 effect where alleles are flipped
    if ("beta.B" %in% names(M)) {
      M[flipped, beta.B := -beta.B]
    }
  }
}

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# Retain harmonisable, non-ambiguous variants
M[(same | flipped) & !ambiguous]

} else {

  warning(
    "Allele columns were not found; harmonisation was skipped."
  )

  M
}
}

# Retrieve genomic window for a PW-GWAS block

get_window <- function(block_data, block_id, pad = PAD) {

  region <- block_data[
    block == block_id,
    .(chr, st, sp)
  ]

  if (!nrow(region)) {
    stop("Block ", block_id, " was not found.")
  }

  data.table(
    chr = region$chr[1],
    st = max(1L, region$st[1] - pad),
    sp = region$sp[1] + pad
  )
}

# -----
# 6. Colocalisation function
# -----

run_coloc_block <- function(
  block_id,
  block_data
) {

  region <- get_window(
    block_data,
    block_id

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)

# Load the relevant genomic region
trait1 <- fread(TRAIT1_FILE)[
  chr == region$chr &
  pos >= region$st &
  pos <= region$sp
]

trait2 <- fread(TRAIT2_FILE)[
  chr == region$chr &
  pos >= region$st &
  pos <= region$sp
]

if (!nrow(trait1) || !nrow(trait2)) {

  return(
    data.table(
      block = block_id,
      note = "No overlapping SNPs in region"
    )
  )
}

# Merge and harmonise
M <- merge_gwas(
  trait1,
  trait2
)

M <- harmonise_gwas(M)

M <- unique(
  M,
  by = "snp"
)

# -----
# Variant-level QC
# -----

if (!"MAF.A" %in% names(M)) {
  M[, MAF.A := 0.5]
}

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}

if (!"p.A" %in% names(M)) {
  M[, p.A := 1]
}

# Retain common variants with evidence of association
# in Trait 1
M <- M[
  MAF.A >= 0.01 &
  p.A < 0.05
]

# Remove variants with invalid effect estimates
M <- M[
  is.finite(beta.A) &
  is.finite(se.A) &
  is.finite(beta.B) &
  is.finite(se.B)
]

# Require sufficient variants for regional inference
if (nrow(M) < 100) {

  return(
    data.table(
      block = block_id,
      n_snps = nrow(M),
      note = "Too few SNPs after QC"
    )
  )
}

# -----
# Define coloc datasets
# -----

# Trait 1: quantitative phenotype
D1 <- list(
  beta      = M$beta.A,
  varbeta   = M$se.A^2,
  snp       = M$snp,
  type      = "quant",
  sdY       = 1

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)

# Trait 2: case-control phenotype
D2 <- list(
  beta      = M$beta.B,
  varbeta   = M$se.B^2,
  snp       = M$snp,
  type      = "cc",
  s         = case_prop,
  N         = N_total
)

# -----
# Run coloc
# -----

coloc_fit <- coloc.abf(
  dataset1 = D1,
  dataset2 = D2,
  p1       = 1e-4,
  p2       = 1e-4,
  p12      = 1e-6
)

summary_stats <- as.list(
  coloc_fit$summary
)

data.table(
  block      = block_id,
  n_snps     = nrow(M),
  PP.H0.abf  = summary_stats$PP.H0.abf,
  PP.H1.abf  = summary_stats$PP.H1.abf,
  PP.H2.abf  = summary_stats$PP.H2.abf,
  PP.H3.abf  = summary_stats$PP.H3.abf,
  PP.H4.abf  = summary_stats$PP.H4.abf
)
}

# -----
# 7. Model 3 regions: PW-GWAS evidence for shared association
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if (length(shared_blocks)) {

  cat(
    "\nRunning coloc for Model 3/shared PW-GWAS blocks...\n"
  )

  results_shared <- rbindlist(
    lapply(
      shared_blocks,
      run_coloc_block,
      block_data = blocks_filtered
    ),
    fill = TRUE
  )

  fwrite(
    results_shared,
    SHARED_OUT,
    sep = "\t"
  )

  cat(
    "Wrote:",
    SHARED_OUT,
    "\n"
  )
} else {

  cat(
    "No high-confidence Model 3 blocks identified.\n"
  )
}

# -----
# 8. Model 4 regions: PW-GWAS evidence for distinct association
# -----

if (length(distinct_blocks)) {

  cat(
    "\nRunning coloc for Model 4/distinct PW-GWAS blocks...\n"
  )

  results_distinct <- rbindlist(
    lapply(
      distinct_blocks,

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        run_coloc_block,
        block_data = blocks_filtered
    ),
    fill = TRUE
)

fwrite(
  results_distinct,
  DISTINCT_OUT,
  sep = "\t"
)

cat(
  "Wrote:",
  DISTINCT_OUT,
  "\n"
)
} else {

  cat(
    "No high-confidence Model 4 blocks identified.\n"
  )
}

# -----
# 9. Completion
# -----

cat("\n-----\n")
cat(
  "Completed PW-GWAS-informed colocalisation for",
  TRAIT_1,
  "vs",
  TRAIT_2,
  "\n"
)

cat(
  "\nInterpretation:\n",
  " PP.H3.abf = evidence that the two traits have distinct causal variants\n",
  " PP.H4.abf = evidence that the two traits share a causal variant\n"
)

cat("-----\n")

```